

Chapter 1

Deterministic and Stochastic Correlation

1.1 Outcomes and Probabilities of Outcomes

1.1.1 Normal Distribution

The **normal distribution** has a probability density function (PDF) of

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[\frac{-(x - \mu)^2}{2\sigma^2} \right]. \quad (1.1)$$

This PDF is valid since it is non-negative for values of support $x \in \mathbb{R}$, and because the integral from $-\infty$ to ∞ is equal to unity. The **cumulative distribution** function (CDF) is

$$P(x) = \int_{-\infty}^x p(\tau) d\tau = \frac{1}{2} \left(1 + \operatorname{erf} \left[\frac{x - \mu}{\sqrt{2\sigma^2}} \right] \right). \quad (1.2)$$

As $x \rightarrow -\infty$, $P \rightarrow 0$. As $x \rightarrow \infty$, $P \rightarrow 1$. The **mean value** is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} p(x) \cdot x dx \quad (1.3)$$

$$= \int_{-\infty}^{\infty} \frac{x}{\sqrt{2\pi\sigma^2}} \exp \left[\frac{-(x - \mu)^2}{2\sigma^2} \right] dx \quad (1.4)$$

$$= \mu. \quad (1.5)$$

The **variance** of x is

$$\text{VAR}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2, \quad (1.6)$$

$$= \int_{-\infty}^{\infty} p(x) \cdot x^2 dx - \mu^2, \quad (1.7)$$

$$= \int_{-\infty}^{\infty} \frac{x^2}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right] dx - \mu^2, \quad (1.8)$$

$$= \sigma^2. \quad (1.9)$$

Remark 1.1.1. For the normal distribution, the mean μ is also equal to the median and mode. The skew and kurtosis are both zero.

Remark 1.1.2. The normal distribution is also called the **Gaussian distribution**. See Stahl¹ for an account for how *normal* arose, and how this naming “... had the disadvantage of leading people to believe that all other distributions of frequency are in one sense or another *abnormal*.”

1.2 Expectation

- Given a series of outcomes, expectation is a **measure** of that series, which will be shaped by the probability of each of the outcomes. Expectation is also called the weighted average.
- In the special case of equally likely probability for each outcome, the expectation will be the mean value, also called the simple average.

1.2.1 Non-Equal Likelihood

Let X be a random variable with a finite number of **outcomes** x_i for $i = 1 \dots n$ with probability p_i . The **expectation** of X , denoted as $\bar{x}$, is defined as

$$\mathbb{E}[X] \triangleq \sum_{i=1}^n p_i x_i = \bar{x}. \quad (1.10)$$

Because the sum of all probabilities is identically equal to unity,

$$\sum_{i=1}^n p_i = 1.0, \quad (1.11)$$

¹Stahl S. *The Evolution of the Normal Distribution*, Mathematics Magazine, 79:2, April 2006.

the expected value is considered the **weighted average**, where each p_i is the **weight**.

Remark 1.2.3. The weighted average is also known as the **first moment**, which is analogous to the **center of mass** of a rigid body. The center of mass analogy applies to both the non-equal likelihood case (non-uniform mass density) and equal likelihood case (uniform mass density).

1.2.2 Equal Likelihood

For the special case where each probability is equally likely,

$$p_1 = p_2 = \cdots = p_n = \frac{1}{n}, \quad (1.12)$$

the expectation of $\mathbb{E}[X]$ has the specialized form defined as μ

$$\mathbb{E}[X] \stackrel{(1.12)}{=} \frac{1}{n} \sum_{i=1}^n x_i \triangleq \mu, \quad (1.13)$$

and is referred to as the **simple average**.

Remark 1.2.4. The expectation $\mathbb{E}[X]$ should not be confused with μ , the simple average of X . The simple average (equal likelihood) is a special case of the expectation (non-equal likelihood).

Remark 1.2.5. When multiple variables are considered, *e.g.*, X and Y , the simple averages are denoted μ_x and μ_y or equivalently, $\bar{x}$ and $\bar{y}$, respectively.

Example 1.

Consider a 6-sided dice, with faces 1 to 6. Consider the dice to be a *fair* dice, so that the probability of rolling a 1 is the same as rolling a 2, and is the same as rolling any of the other remaining numbers 3 to 6. Thus, the fair dice has equal likelihood of any of the 6 outcomes, and Eq. (1.13) is used. The expected value is calculated as

$$\mathbb{E}[X] = \mu = \mu_x = \bar{x} = \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) = 3.5 \quad (1.14)$$

...

The 6-sided dice roll in the preceding example can be illustrated through numerical

simulation, implemented in Python below. The results from the Python script are shown in Fig. 1.1.

```

1  #!/usr/bin/env python3
2  # dice_roll.py
3  import matplotlib.pyplot as plt
4  from matplotlib import rc
5  import numpy as np
6  from random import randint # random integer
7  rc('font', **{'family': 'serif', 'serif': ['Computer Modern Roman']})
8  rc('text', usetex=True)
9
10 n_rolls = 40 # number of dice rolls
11 roll_number = np.arange(1, n_rolls + 1)
12 roll_history = np.zeros(n_rolls) # each roll result recorded in time
13 average = np.zeros(n_rolls) # initialize the average
14 # then update after each dice roll
15
16 for roll in roll_number:
17     # roll a fair dice, values between 1 and 6, inclusive
18     roll_outcome = randint(1, 6)
19     print(f'Dice roll number {roll} outcome is {roll_outcome}.')
20     roll_history[roll-1] = int(roll_outcome)
21     sum_after_roll = sum(i for i in roll_history)
22     average[roll-1] = sum_after_roll / roll
23
24 fig = plt.figure(figsize=(8, 4)) # 8 inches wide, 4 inches tall
25
26 ax = fig.add_subplot(1, 1, 1)
27 ax.plot(roll_number, roll_history, 's', color='blue', label='roll outcome')
28 ax.plot(roll_number, average, color='red', marker='o',
29         alpha=0.5, label='roll average')
30 ax.grid()
31 ax.set_xlabel('roll number')
32 ax.set_ylabel('roll outcome and cumulative simple average')
33 ax.set_ylim(0, 7)
34 ax.legend(loc='upper right')
35 plt.show()
36 figure_name = 'dice_roll'
37 fig.savefig(figure_name + '.pdf', bbox_inches='tight')

```

The first moment analog to the center of mass can be illustrated through graph-

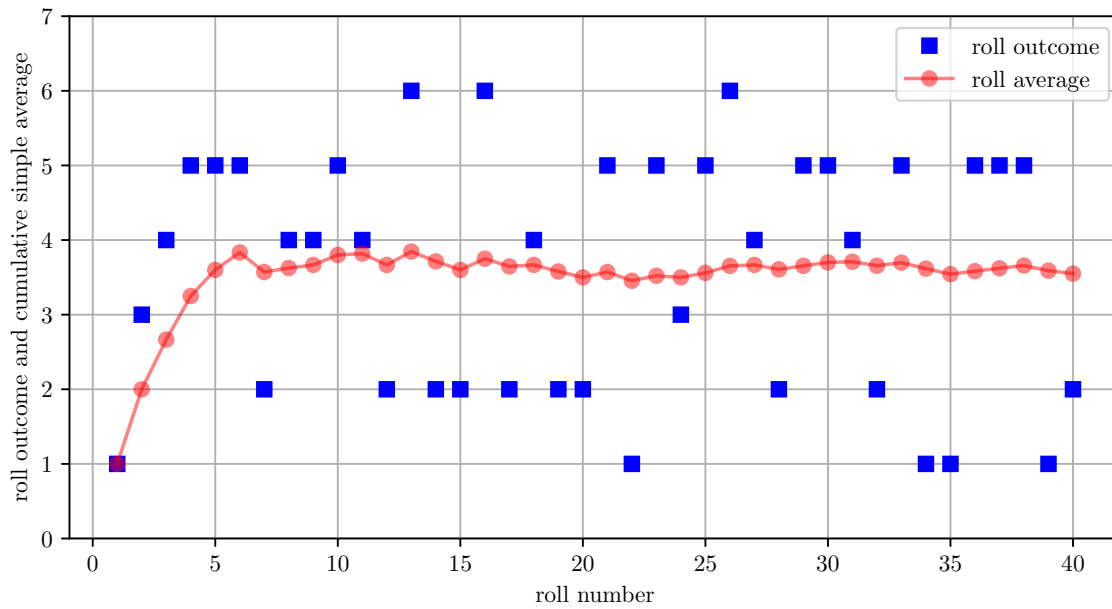


Figure 1.1: Numerical simulation of 40 dice rolls, showing value of each roll and average for all rolls. As the number of rolls increases, the average converges to 3.5, the dice roll expected value.

ical inspection. Fig. 1.2 shows the roll outcomes, $1, \dots, 6$, along the x -axis and the probability of any of the six outcomes, $p_i = 1/6$, normalized to unity (since all of the six probabilities are equal) on the y -axis.

The center of mass of the six blue squares is readily seen at the $x = 3.5$ coordinate, indicated with a red circle. The center of mass is the simple average. This illustrates the equal probability special case, interpreted in the mechanical analog as the uniform density case.

For the general case with non-equal probability, *e.g.*, an unfair or weighted dice, the higher (or lower) probability of a particular outcome is analogous to an increase (or decrease) of local density near the particular x -coordinate of that outcome, and hence as the non-uniform density case.

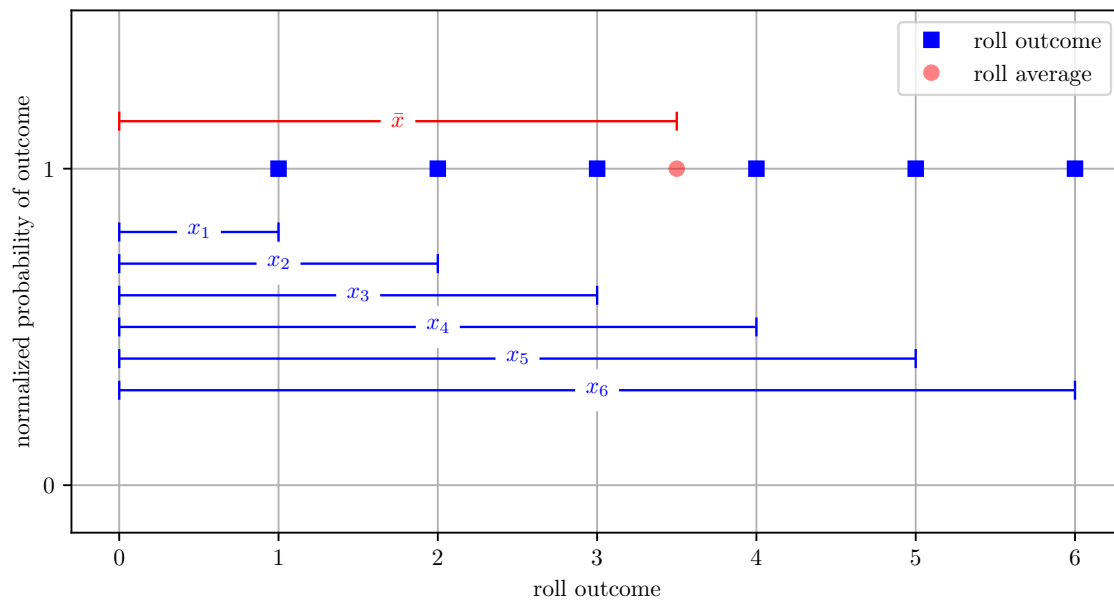


Figure 1.2: Graphical interpretation of the simple average as the first moment center of mass $\bar{x}$.

1.3 Variance

1.3.1 Non-Equal Likelihood

The **Variance** of X is defined as

$$\text{VAR}(X) \triangleq \mathbb{E} [(X - \mathbb{E}[X])^2], \quad (1.15)$$

$$\begin{aligned} & \mathbb{E} [(X - \mathbb{E}[X]) (X - \mathbb{E}[X])], \\ & \mathbb{E} [X^2 - 2 \mathbb{E}[X] X + \mathbb{E}[X] \mathbb{E}[X]], \\ & \mathbb{E}[X^2] - 2 \mathbb{E}[X] \mathbb{E}[X] + \mathbb{E}[X] \mathbb{E}[X], \\ & \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \end{aligned} \quad (1.16)$$

1.3.2 Equal Likelihood

For the special case where each probability is equally likely, (1.16) takes the form

$$\text{VAR}(X) \stackrel{(1.16)}{=} \sum_{i=1}^n p_i x_i^2 - \bar{x}^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 \triangleq \sigma_x^2, \quad (1.17)$$

and is referred to as the **simple variance**.

Remark 1.3.6. The form of (1.17) is often stated in slightly different form,

$$\sigma_x^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - 2\bar{x}^2 + \bar{x}^2, \quad (1.18)$$

$$\stackrel{(1.13)}{=} \frac{1}{n} \sum_{i=1}^n x_i^2 - \frac{2\bar{x}}{n} \sum_{i=1}^n x_i + \frac{n\bar{x}^2}{n}, \quad (1.19)$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i^2 - 2\bar{x}x_i + \bar{x}^2), \quad (1.20)$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2. \quad (1.21)$$

1.4 Covariance

1.4.1 Non-Equal Likelihood

For two jointly distributed real-valued random variables X and Y , with **expected values**² $\mathbb{E}[X]$ and $\mathbb{E}[Y]$, respectively, the **covariance** is defined as

$$\text{Cov}(X, Y) \triangleq \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])], \quad (1.22)$$

$$\begin{aligned} & \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] - \mathbb{E}[X] \mathbb{E}[Y] + \mathbb{E}[X] \mathbb{E}[Y], \\ & \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]. \end{aligned} \quad (1.23)$$

Let the random variable pair (X, Y) take on values (x_i, y_i) for $i = 1, \dots, n$ with probability p_i . The covariance is defined as

$$\text{Cov}(X, Y) \triangleq \sum_{i=1}^n p_i (x_i - \mathbb{E}[X]) (y_i - \mathbb{E}[Y]). \quad (1.24)$$

Remark 1.4.7. The covariance of a random variable with itself is its variance,

$$\text{Cov}(X, X) = \text{VAR}(X), \quad (1.25)$$

which is seen by comparing (1.22) to (1.15).

1.4.2 Equal Likelihood

If the probability of each outcome is equal, the $p_i = 1/n$, and (1.24) becomes

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) (y_i - \bar{y}). \quad (1.26)$$

1.5 Correlation

1.5.1 Pearson Correlation Coefficient

The **Pearson correlation coefficient** (PCC)³ is a measure of the linear correlation between two variables X and Y .

²Also known as **mean values**.

³Also known as the Pearson's r coefficient, Pearson product-moment correlation coefficient (PPMCC), and the bivariate correlation.

When a pair of random variables (X, Y) are applied to a population, the PCC is called the **population correlation coefficient** and denoted as $\rho_{X,Y}$ and is defined as

$$\rho_{X,Y} \triangleq \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad (1.27)$$

where

- Cov is the covariance of X and Y ,
- σ_X is the standard deviation of X , and
- σ_Y is the standard deviation of Y .

The PCC is bounded as

$$-1 \leq \rho_{X,Y} \leq 1. \quad (1.28)$$

- A value of -1 indicates complete negative linear correlation.
- A value of 0 indicates the absence of linear correlation.
- A value of +1 indicates complete positive linear correlation.

Remark 1.5.8. When $\mathbb{E}[XY] \approx \mathbb{E}[X] \mathbb{E}[Y]$, then the $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]$ is prone to so-called “catastrophic cancellation” with floating point calculations. A numerically stable algorithm should be used.

1.6 Linear Regression

1.6.1 Univariate Linear Regression

Model

Univariate linear regression is a linear relationship between an independent variable $x \in \mathbb{R}$, also called the *explanatory variable*, and a dependent variable $y \in \mathbb{R}$, also called a *response variable*.

The **model function** is

$$y = wx + b \tag{1.29}$$

where

- $w \in \mathbb{R}$ is the slope (also called **weight**) acting on the independent variable, and
- $b \in \mathbb{R}$ is the intercept (also called **bias**).

The two real-valued numbers (w, b) are called **model parameters**.

Data

Assume n pair of observations (x_i, y_i) , $i = 1 \dots n$, have been made. Assume equal likelihood of each observation. Given the observations, the objective of univariate linear regression is to

1. find **best fit model parameters** $(\hat{w}, \hat{b})$, and
2. quantify the **correlation coefficient**, which is an objective measure of how well the best fit model describes the underlying observed data.

Error Function

For each observation (x_i, y_i) pair, there is an error $e_i(w, b)$ that is defined as the difference between the actual observed response y_i and the model's predicted response $w x_i + b$, defined as

$$e_i(w, b) \triangleq y_i - (w x_i + b). \tag{1.30}$$

The error accumulated for all n observations can be written as

$$\| \mathbf{e}(w, b) \|_1 \triangleq \sum_{i=1}^n |e_i(w, b)|. \tag{1.31}$$

This is the L1-norm of the error vector $\mathbf{e} = [e_1, e_2, \dots, e_n] \in \mathbb{R}^n$. See Section 1.7.4 for more details on p-norms.

The model parameters (w, b) are said to be best fit when the error is minimized. While the L1-norm error in (1.30) may be used as the objective error function, the L2-norm is frequently used as an alternative to the L1-norm.

$$\|\mathbf{e}(w, b)\|_2 \triangleq \sum_{i=1}^n (e_i(w, b))^2 = \sum_{i=1}^n (y_i - (wx_i + b))^2. \quad (1.32)$$

Remark 1.6.9. The use of the L2-norm is called **ordinary least squares** (OLS), wherein the goal is to minimize the square of the sum of the deviations of the observation output y_i and the model's output $(wx_i + b)$. This is also called the RSS for **residual sum of squares**.

Remark 1.6.10. The TSS, **total sum of squares**, gives rise to the **R-squared metric**, defined as

$$R^2 \triangleq 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\sum_{i=1}^n (e_i(w, b))^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (1.33)$$

The square of the Pearson Correlation Coefficient (PCC) is equal to the R-squared value,

$$(\rho_{X,Y})^2 = R^2. \quad (1.34)$$

Result 1.6.1 Given a univariate linear regression model and data consisting of n pair of observations (x_i, y_i) , $i = 1 \dots n$, the best fit model parameters $(\hat{w}, \hat{b})$ are

$$\hat{w} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}. \quad (1.35)$$

and

$$\hat{b} = \bar{y} - \hat{w} \bar{x}. \quad (1.36)$$

Proof.

For this model, start with the L2-error norm, defined as

$$\|e(w, b)\|_2 \triangleq \sum_{i=1}^n (e_i(w, b))^2 = \sum_{i=1}^n (y_i - wx_i - b)^2. \quad (1.37)$$

Using the notation $E_2 \in \mathbb{R}$ to denote the L2-error norm in (1.32), we expand and simplify the squared term,

$$E_2(w, b) = \sum_{i=1}^n y_i^2 - 2y_iwx_i - 2y_ib + w^2x_i^2 + 2wx_ib + b^2. \quad (1.38)$$

Since E_2 is the summation of squared quantities, $E_2(w, b) \geq 0 \forall w \in \mathbb{R}, b \in \mathbb{R}$. $E_2(w, b)$ behaves as a quadratic, scalar function. Values of (w, b) that minimize the error E_2 are found by taking partial derivatives of the error function with respect to the parameters w and b :

$$\frac{\partial E_2}{\partial w} = \sum_{i=1}^n -2y_ix_i + 2wx_i^2 + 2x_ib \quad (1.39)$$

and

$$\frac{\partial E_2}{\partial b} = \sum_{i=1}^n -2y_i + 2wx_i + 2b. \quad (1.40)$$

Setting (1.40) equal to zero, and using the definitions of $\bar{x}$ and $\bar{y}$ from (1.13),

$$\hat{b} = \bar{y} - w\bar{x} \text{ to minimize } E_2. \quad (1.41)$$

Substituting (1.41) into (1.38) for values of $\hat{b}$ that minimize E_2 allows (1.38) to be written as a function of w alone,

$$E_2(w) = \sum_{i=1}^n y_i^2 - 2y_iwx_i - 2y_i(\bar{y} - w\bar{x}) + w^2x_i^2 + 2wx_i(\bar{y} - w\bar{x}) + (\bar{y} - w\bar{x})^2. \quad (1.42)$$

Then,

$$\frac{dE_2}{dw} = \sum_{i=1}^n -2y_ix_i + 2y_i\bar{x} + 2wx_i^2 + 2x_i\bar{y} - 4wx_i\bar{x} + 2(\bar{y} - w\bar{x})(-\bar{x}) \stackrel{\text{set}}{=} 0, \quad (1.43)$$

or

$$0 = \sum_{i=1}^n -y_ix_i + y_i\bar{x} + wx_i^2 + x_i\bar{y} - 2wx_i\bar{x} + w\bar{x}\bar{x} - \bar{x}\bar{y}, \quad (1.44)$$

$$0 = \sum_{i=1}^n -(x_i - \bar{x})(y_i - \bar{y}) + w(x_i - \bar{x})^2. \quad (1.45)$$

Solving for w is then the best fit $\hat{w}$

$$\hat{w} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}. \quad (1.46)$$

Multiplying both the numerator and the denominator by $\frac{1}{n}$ gives

$$\hat{w} = \frac{\text{Cov}(X, Y)}{\text{VAR}(X)}. \quad (1.47)$$

Thus the first objective of univariate linear regression, find model parameters $(\hat{w}, \hat{b})$ the best fit to the observed data (by minimizing the error between the model and the data) has been accomplished. ■

The second and final objective, to quantify the quality of the best fit, follows.

Result 1.6.2

$$\rho_{X,Y} = \frac{\hat{w} \sigma_x}{\sigma_y}. \quad (1.48)$$

Proof.

Using the sample correlation coefficient $\rho_{X,Y}$, the sample (uncorrected) standard deviation σ_x and σ_y , the sample variance and covariance,

$$\text{VAR}(X) = \sigma_x^2, \quad (1.49)$$

$$\text{COV}(X, Y) = \rho_{X,Y} \sigma_x \sigma_y, \quad (1.50)$$

then

$$\hat{w} = \frac{\rho_{X,Y} \sigma_y}{\sigma_x} \implies \rho_{X,Y} = \frac{\hat{w} \sigma_x}{\sigma_y}. \quad (1.51)$$

■

Result 1.6.3 As long as the univariate linear regression model includes the intercept (bias) term b (*i.e.*, the model is not forced to go through the origin by virtue of $b \stackrel{\text{set}}{=} 0$), then

1. the sum of the errors is zero

$$\sum_{i=1}^n e_i = 0, \quad \text{and} \quad (1.52)$$

2. the model regression line goes through the center of mass point $(\bar{x}, \bar{y})$.

Proof.

For part 1, start from the definition (1.30),

$$e_i = y_i - wx_i - b, \quad (1.53)$$

add up all the errors, $i = 1 \dots n$, and divide by n ,

$$\frac{1}{n} \sum_{i=1}^n e_i = \frac{1}{n} \sum_{i=1}^n y_i - wx_i - b. \quad (1.54)$$

Substitute the definitions of $\bar{x}$ and $\bar{y}$ from (1.13)

$$= \bar{y} - w\bar{x} - b \quad (1.55)$$

and finally, substitute the model equation (1.29) with values of $y \mapsto \bar{y}$ and $x \mapsto \bar{x}$,

$$= \bar{y} - w\bar{x} - \bar{y} + w\bar{x} = 0 \quad (1.56)$$

which is the result for part 1. For part 2, show that the error at the point $(\bar{x}, \bar{y})$ is zero,

$$e_i|_{(\bar{x}, \bar{y})} = y_i|_{\bar{y}} - (w x_i|_{\bar{x}} + b) = \frac{1}{n} \sum_{i=1}^n y_i - \left(\frac{w}{n} \sum_{i=1}^n x_i + b \right). \quad (1.57)$$

Rearranging,

$$e_i|_{(\bar{x}, \bar{y})} = \frac{1}{n} \sum_{i=1}^n (y_i - (wx_i + b)) = \frac{1}{n} \sum_{i=1}^n e_i = 0 \quad (1.58)$$

by the result in part 1. This shows that the point $(\bar{x}, \bar{y})$ is on the line described by the model in (1.29). ■

Example 2.

Here we present a small example of artificially constructed data useful for purposes of illustration and unit testing.⁴ Consider the following $n = 5$ pair of observations (x_i, y_i) , $i = 1 \dots n$,

$$\mathbf{x} = [1.00, 2.00, 3.00, 4.00, 5.00] \quad (1.59)$$

$$\mathbf{y} = [1.00, 2.00, 1.30, 3.75, 2.25] \quad (1.60)$$

Calculating the averages,

$$\bar{x} = (1.00 + 2.00 + 3.00 + 4.00 + 5.00)/5 = 3.00 \quad (1.61)$$

$$\bar{y} = (1.00 + 2.00 + 1.30 + 3.75 + 2.25)/5 = 2.06 \quad (1.62)$$

⁴Adapted from <http://onlinestatbook.com/2/regression/intro.html>

Next we calculate the population variances and covariance. Technically, we should use the *sample* covariance and *sample* variance in place of the *population* formulae. The difference in the factor of $1/n$ for the population and $1/(n - 1)$ for the sample appear in both the numerator and denominator for the calculations of $\hat{w}$, $\hat{b}$ and $\rho_{X,Y}$, and thus they cancel out, making the calculations independent of the use of sample or population measures.

$$\begin{aligned}\text{VAR}(\mathbf{x}) = & ((1.00 - 3.00)^2 + \\ & (2.00 - 3.00)^2 + \\ & (3.00 - 3.00)^2 + \\ & (4.00 - 3.00)^2 + \\ & (5.00 - 3.00)^2) / 5 = 2.00\end{aligned}\tag{1.63}$$

$$\begin{aligned}\text{VAR}(\mathbf{y}) = & ((1.00 - 2.06)^2 + \\ & (2.00 - 2.06)^2 + \\ & (1.30 - 2.06)^2 + \\ & (3.75 - 2.06)^2 + \\ & (2.25 - 2.06)^2) / 5 = 0.9194\end{aligned}\tag{1.64}$$

$$\begin{aligned}\text{COV}(\mathbf{x}, \mathbf{y}) = & ((1.00 - 3.00)(1.00 - 2.06) + \\ & (2.00 - 3.00)(2.00 - 2.06) + \\ & (3.00 - 3.00)(1.30 - 2.06) + \\ & (4.00 - 3.00)(3.75 - 2.06) + \\ & (5.00 - 3.00)(2.25 - 2.06)) / 5 = 0.850\end{aligned}\tag{1.65}$$

Then, with (1.35), or equivalently (1.47),

$$\hat{w} = \frac{\text{COV}(\mathbf{x}, \mathbf{y})}{\text{VAR}(\mathbf{x})} = \frac{0.850}{2.00} = 0.425\tag{1.66}$$

as the slope (weight), and with (1.36)

$$\hat{b} = \bar{y} - \hat{w}\bar{x} = 2.06 - 0.425 * 3.00 = 0.785\tag{1.67}$$

as the intercept (bias). The Pearson correlation coefficient (PCC) is found with (1.27) as,

$$\begin{aligned}\rho_{X,Y} &= \frac{\hat{w}\sigma_x}{\sigma_y} = \frac{\hat{w}\sqrt{\text{VAR}(\mathbf{x})}}{\sqrt{\text{VAR}(\mathbf{y})}} = \frac{\text{COV}(\mathbf{x}, \mathbf{y})}{\sqrt{\text{VAR}(\mathbf{x})}\sqrt{\text{VAR}(\mathbf{y})}} \\ &= \frac{0.425 * \sqrt{2.00}}{\sqrt{0.9194}} = \frac{0.850}{\sqrt{2.00}\sqrt{0.9194}} = 0.627\end{aligned}\tag{1.68}$$

...

Following is a Python implementation of this example, and the resulting graphical output, showing the observed data, the mean and variance values, and the linear regression relationship.

```

1  #!/usr/bin/env python3
2  # lin_regress_1.py
3  import numpy as np
4  import matplotlib as mpl
5  from matplotlib import rc
6  import matplotlib.pyplot as plt
7  from matplotlib.ticker import AutoMinorLocator, MultipleLocator, FuncFormatter
8  rc('font', **{'family': 'serif', 'serif': ['Computer Modern Roman']})
9  rc('text', usetex=True)
10
11  lw = 1 # line width for leader lines
12
13  x = np.array([1.00, 2.00, 3.00, 4.00, 5.00])
14  y = np.array([1.00, 2.00, 1.30, 3.75, 2.25])
15  theta = np.linspace(0, 2*np.pi, 100)
16
17  x_ave = np.mean(x)
18  y_ave = np.mean(y)
19
20  x_var = np.var(x)
21  y_var = np.var(y)
22

```

```

23  # the [0,1] component of the covariance matrix
24  xy_cov = np.cov(x, y, bias=True)[0,1]
25
26  w_hat = xy_cov / x_var
27  b_hat = y_ave - w_hat * x_ave
28  x_lin = np.linspace(0, 6, 2)
29  y_lin = w_hat * x_lin + b_hat  # y = m x + b
30
31  dx = dy = 0.15
32  xmax = 6
33  ymax = 6
34  fig = plt.figure(figsize=(xmax, ymax))  # (width, height) inches
35  ax = fig.add_subplot(1, 1, 1)
36  ax.plot(x, y, 's', color='blue', label='observation')
37  ax.plot(x_ave, y_ave, 'o', color='red', alpha=0.5,
38          label=r'($\bar{x}, \bar{y}) = (' +
39              '{:.2f}'.format(x_ave) + ', ' +
40              '{:.2f}'.format(y_ave) + ')')
41  ax.plot(x_var * np.cos(theta) + x_ave,
42          y_var * np.sin(theta) + y_ave,
43          color='green', linewidth=1.0,
44          label=r'$\sigma_x^2, \sigma_y^2 = ' +
45              '{:.2f}'.format(x_var) + ', ' +
46              '{:.4f}'.format(y_var))
47  ax.plot(x_lin, y_lin, color='red', linewidth=1,
48          label=r'$y = \hat{w} x + \hat{b} = $ ' +
49              '{:.3f}'.format(w_hat) + ' $x$; + $ ' +
50              '{:.3f}'.format(b_hat) )
51  ax.axis('equal')
52  ax.grid()
53  ax.set_xlabel('explanatory variable $x$')
54  ax.set_ylabel('response variable $y$')
55  ax.set_xlim(0 - dx, xmax + dx)
56  ax.set_ylim(0 - dy, ymax + dy)
57  ax.legend(loc='upper left')
58
59  plt.show()
60
61  figure_name = 'lin_regress_1'
62  fig.savefig(figure_name + '.pdf', bbox_inches='tight')

```

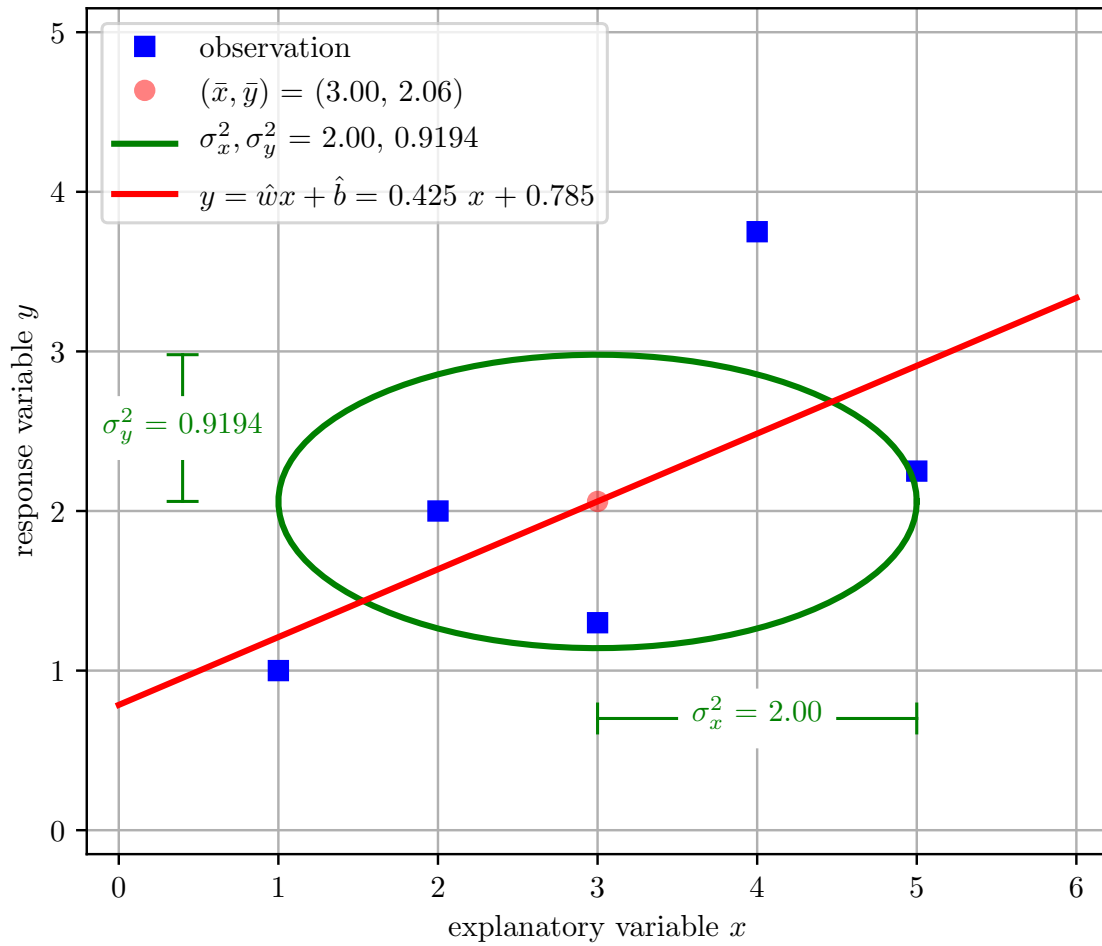


Figure 1.3: Observed data points (blue), the mean values $\bar{x}$ and $\bar{y}$, the variance ellipse, and the linear regression relationship.

1.7 Cross-Correlation (Series Dot Product)

The cross-correlation is a measure of the *similarity in time* of two time series signals. It is often referred to as the *dot product* of two time series because the cross-correlation operation is a dot production operation.

1.7.1 Vector Dot Product

Consider two vectors $\mathbf{a}$ and $\mathbf{b}$ in three-dimensional space, measured from the same origin and with the same coordinate system, with components,

$$\mathbf{a} = [a_1, a_2, a_3], \quad (1.69)$$

$$\mathbf{b} = [b_1, b_2, b_3]. \quad (1.70)$$

The dot product of the two vectors is simply,

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i = a_1 b_1 + a_2 b_2 + a_3 b_3 \quad (1.71)$$

$$\text{or generally} = \sum_{i=1}^n a_i b_i. \quad (1.72)$$

Next, consider $\mathbf{a}(t)$ and $\mathbf{b}(t)$ be two time-varying signals, occurring over time interval $[t_0, t_f]$. Let the amplitudes of $\mathbf{a}(t)$ and $\mathbf{b}(t)$ be measured from zero to t_f at n equally distributed points such that the signals are sampled at n uniform discrete times in the interval. The two vectors are written explicitly as

$$\mathbf{a} = [a_0, a_1, a_2, \dots, a_{n-3}, a_{n-2}, a_{n-1}], \quad (1.73)$$

$$\mathbf{b} = [b_0, b_1, b_2, \dots, b_{n-3}, b_{n-2}, b_{n-1}]. \quad (1.74)$$

Remark 1.7.11. To account for the *initial condition* of the time series, we decrement the counter,

$$1 \dots n, \quad (1.75)$$

to become

$$0 \dots n - 1. \quad (1.76)$$

In both of these cases, the number of sample points is identical, but the indexing is different by one.

Remark 1.7.12. Notice that the two signals $\mathbf{a}(t)$ and $\mathbf{b}(t)$ are *synchronized* in time such that

$$\mathbf{t} = [t_0, t_1, t_2, \dots, t_{n-3}, t_{n-2}, t_{n-1}]. \quad (1.77)$$

Notice that $\Delta t = t_1 - t_0 = t_2 - t_1 = \dots t_{n-1} - t_{n-2}$ and that the signals are sampled at a frequency of $\frac{1}{\Delta t}$ Hz.

1.7.2 Series Dot Product

The dot product of these two signals is

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i = a_0 b_0 + a_1 b_1 + a_2 b_2 + \dots + a_{n-1} b_{n-1} \quad (1.78)$$

$$\text{or generally} = \sum_{i=0}^{n-1} a_i b_i. \quad (1.79)$$

For comparison of two time series, the dot product will be referred to as the **cross-correlation score**.

Example 3.

This example is a reproduction of the cross-correlation example on the Anomaly site.⁵ Let vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ be defined as follows,

$$\mathbf{a} = [1, 2, -2, 4, 2, 3, 1, 0] \quad (1.80)$$

$$\mathbf{b} = [2, 3, -2, 3, 2, 4, 1, -1] \quad (1.81)$$

$$\mathbf{c} = [-2, 0, 4, 0, 1, 1, 0, -2] \quad (1.82)$$

Then the cross-correlation score between $\mathbf{a}$ and $\mathbf{b}$ is found to be high,

$$\mathbf{a} \cdot \mathbf{b} = 41; \quad (1.83)$$

whereas, the cross-correlation score between $\mathbf{a}$ and $\mathbf{c}$ is found to be low,

$$\mathbf{a} \cdot \mathbf{c} = -5. \quad (1.84)$$

...

⁵See <https://anomaly.io/understand-auto-cross-correlation-normalized-shift/>

1.7.3 Limitation of Cross-Correlation

A significant limitation of the cross-correlation method is its dependence on *amplitude*:

- Large positive magnitudes increase the cross-correlation.
- Large negative magnitudes decrease the cross-correlation.
- A cross-correlation score shouldn't get better (positive) or worse (negative) if the amplitude of the signal is increased or decreased.
- Shape similarity is obfuscated by amplitude dependence. For example, the three cross-correlation scores generated from comparison of three signals, $A \sin(t)$, $B \sin(t)$, and $C \sin(t)$, all of the same shape, $\sin(t)$, but with unique amplitudes A, B, C , will report that two out of the three signals are better correlated by virtue of their closer amplitudes. In a sense, this result is satisfactory, since amplitude distinguishes one signal from another. However, this result is incomplete, since the amplitude dependence obfuscates the fact that all three signals have the same basic shape.
- Shape similarity obfuscation is easily seen with shape *self*-similarity, that is, using the *same* signal for both vectors in the cross-correlation. In this case

$$\mathbf{a} \cdot \mathbf{a} > \mathbf{a} \cdot \frac{\mathbf{a}}{2}, \quad (1.85)$$

which is a sensible outcome, but not helpful to distinguish shape similarity.

To remedy the shortcoming of cross-correlation caused by amplitude dependence, **vector normalization** is employed, as explained in a following section. Prior to this, however, we make a brief introduction to the p-norm.

1.7.4 p-norm

For any any number $p \in \mathbb{R}$ and $p \geq 1$, the **p-norm** is defined as

$$\|\mathbf{a}\|_p \triangleq \left(\sum_{i=0}^{n-1} |a_i|^p \right)^{\frac{1}{p}}. \quad (1.86)$$

Example 4.

The L2-norm, also called the **Euclidean norm**, is defined as

$$\|\mathbf{a}\|_2 \triangleq \left(\sum_{i=0}^{n-1} |a_i|^2 \right)^{\frac{1}{2}}, \quad (1.87)$$

and has the physical interpretation as the length of the vector on which the L2-norm operates.

The L1-norm, also called the **Manhattan taxicab norm**, is defined as

$$\| \mathbf{a} \|_1 \triangleq \left(\sum_{i=0}^{n-1} | a_i | \right), \quad (1.88)$$

and has the physical interpretation as the cumulative length (mileage) accumulated for each segment a_i .

The **absolute value** is the L1-norm on a one-dimensional vector space,

$$\| \mathbf{a} \|_1 \triangleq \left(\sum_{i=0}^0 | a_i | \right) = | a_0 |, \quad (1.89)$$

...

1.7.5 Normalized Cross-Correlation

When vector normalization is used with cross-correlation, the resulting method is referred to as *normalized cross-correlation*. Normalized cross-correlation is the same as cross-correlation with one exception: The vectors used in the dot product are normalized by their respective L2-norm.

The normalized cross-correlation for the three-dimensional vectors $\mathbf{a}$ and $\mathbf{b}$ is defined as

$$\frac{\mathbf{a}}{\| \mathbf{a} \|_2} \cdot \frac{\mathbf{b}}{\| \mathbf{b} \|_2} \triangleq \frac{a_i}{\sqrt{\sum_{i=1}^n a_i a_i}} \cdot \frac{b_i}{\sqrt{\sum_{i=1}^n b_i b_i}} \quad (1.90)$$

$$= \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\left(\sqrt{a_1^2 + a_2^2 + a_3^2} \right) \left(\sqrt{b_1^2 + b_2^2 + b_3^2} \right)}. \quad (1.91)$$

The foregoing result generalizes to time series $\mathbf{a}(t)$ and $\mathbf{b}(t)$ as

$$\frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \frac{\mathbf{b}}{\|\mathbf{b}\|} \triangleq \frac{\sum_{i=0}^{n-1} a_i b_i}{\left(\sqrt{\sum_{i=0}^{n-1} a_i^2} \right) \left(\sqrt{\sum_{i=0}^{n-1} b_i^2} \right)} \quad (1.92)$$

Consider again the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ from Example 1.

Example 5.

Consider again the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ from Example 1. The *normalized* cross-correlation score between $\mathbf{a}$ and $\mathbf{b}$ is found to be high (relative to a maximum of 1.0),

$$\frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \frac{\mathbf{b}}{\|\mathbf{b}\|} = \frac{41}{\sqrt{39} \sqrt{48}} = 0.947; \quad (1.93)$$

whereas, the *normalized* cross-correlation score between $\mathbf{a}$ and $\mathbf{c}$ is found to be low (relative to no correlation for 0.0),

$$\frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \frac{\mathbf{c}}{\|\mathbf{c}\|} = \frac{-5}{\sqrt{39} \sqrt{26}} = -0.157. \quad (1.94)$$

...

The **benefits** of *normalized* cross-correlation are two-fold: Amplitude independence and unit normal scoring. Amplitude independence is seen as

$$\mathbf{a} \cdot \mathbf{a} = \mathbf{a} \cdot \frac{\mathbf{a}}{2}. \quad (1.95)$$

The normalized cross-correlation score now ranges between -1 and +1.

- +1 is perfect correlation,
- 0 is no correlation, and
- -1 is anti-correlation (correlated, but in opposite phase).

1.7.6 Normalized Cross-Correlation with Time Shift

The motivation for **normalization** was to remove amplitude effects from the cross-correlation score. These effects are seen along the y -axis. A complementary effect, called **phase effect**, can exist along the x -axis.

Specifically, two time series may be highly correlated, or even identical, but if out of phase, will cause the cross-correlation and the normalized cross-correlation to report a score of zero.

To remedy this outcome, a second strategy can be formulated. In previous sections, the dot product and the normalized dot product were used for cross-correlation and normalized cross-correlation, respectively. Now, we will shift one of the two signals incrementally in time to both the left and the right, to create what is referred to as the **sliding dot product** and **sliding normalized dot product**, respectively.

Conceptually, the sliding dot product strategy computes the dot product between two time series for all combination of a_i and b_i to determine if a **phase lead** or **phase lag** results in a better correlation score.

The result of the sliding dot product is an expansion of inner products, called **cross-correlation coefficients**, which will be labeled as vector $\mathbf{c}$, with dimension $2n - 1$, generated by the sliding mechanism. The sliding dot product is illustrated in matrix form, where time series $\mathbf{b}$ slides forward and backward incrementally in time, relative to time series $\mathbf{a}$. This operation takes the explicit form,

$$\begin{array}{c}
 \text{initial condition, len} = 1 \\
 \hline
 \text{slide-forward} \\
 \text{operation} \\
 \vdots \\
 \text{len} = n - 1 \\
 \hline
 \text{slide-backward} \\
 \text{operation} \\
 \vdots \\
 \text{len} = n - 1
 \end{array}
 \left\{ \begin{array}{c}
 c_0 \\
 c_1 \\
 c_2 \\
 \vdots \\
 c_{n-1} \\
 \hline
 c_{-1} \\
 c_{-2} \\
 \vdots \\
 c_{-(n-1)}
 \end{array} \right\} = \quad (2n-1 \times 1)$$

$$\begin{bmatrix}
b_0 & b_1 & b_2 & \cdots & b_{n-3} & b_{n-2} & b_{n-1} \\
0 & b_0 & b_1 & b_2 & \cdots & b_{n-3} & b_{n-2} \\
0 & 0 & b_0 & b_1 & b_2 & \cdots & b_{n-3} \\
& & & \ddots & & & \vdots \\
& & & & \ddots & & \vdots \\
& & & & & \ddots & \vdots \\
0 & 0 & 0 & 0 & 0 & \cdots & b_0 \\
b_1 & b_2 & \cdots & b_{n-3} & b_{n-2} & b_{n-1} & 0 \\
b_2 & \cdots & b_{n-3} & b_{n-2} & b_{n-1} & 0 & 0 \\
\vdots & & & \ddots & & & \\
\vdots & & & & \ddots & & \\
\vdots & \ddots & & & & & \\
b_{n-1} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}_{(2n-1 \times n)} \quad \left\{ \begin{array}{c} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_{n-3} \\ a_{n-2} \\ a_{n-1} \end{array} \right\}_{(n \times 1)} \quad (1.96)$$

The foregoing matrix form can be reordered row-wise, from smallest to largest value of j , and coded in index form

$$c_j = \sum_{i=0}^{n-1} b_{i-j} a_i, \quad \text{with } j = -(n-1) \dots (n-1). \quad (1.97)$$

The following table provides five samples points for j verify the formula in (1.97).

Table 1.1: Five-point verification of sliding dot product form (1.97).

j	c
$-(n-1)$	$\sum_{i=0}^{n-1} b_{i+(n-1)} a_i = b_{n-1} a_0$
$\dots$	$\dots$
-1	$\sum_{i=0}^{n-1} b_{i+1} a_i = b_1 a_0 + b_2 a_1 + \cdots + b_{n-1} a_{n-2}$
0	$\sum_{i=0}^{n-1} b_i a_i = b_0 a_0 + b_1 a_1 + \cdots + b_{n-1} a_{n-1}$
1	$\sum_{i=0}^{n-1} b_{i-1} a_i = b_0 a_1 + b_1 a_2 + \cdots + b_{n-2} a_{n-1}$
$\dots$	$\dots$
$(n-1)$	$\sum_{i=0}^{n-1} b_{i-(n-1)} a_i = b_0 a_{n-1}$

Example 6.

This example is a reproduction of the normalized cross-correlation with time shift example from the Anomaly site.⁶ Let vectors $\mathbf{a}$ and $\mathbf{b}$ be defined as follows,

$$\mathbf{a} = [0, 1, 2, 3, 4, \quad 0, 1, 2, 3, 4, \quad 0, 1, 2, 3, 4, \quad 0, 1, 2, 3, 4] \quad (1.98)$$

$$\mathbf{b} = [1, 2, 3, 3, 0, \quad 1, 2, 3, 4, 0, \quad 1, 1, 4, 4, 0, \quad 1, 2, 3, 4, 0] \quad (1.99)$$

where slight spacing has been interjected between every fifth element, to help illustrate the pattern embedded in the series. Since the dimension, n , of the signal is 20, there are $2n - 1 = 39$ correlation coefficients, c_i , that can be calculated from the normalized cross-correlation with time shift operation. Below are 9 of the 39 correlation coefficients, showing the initial condition, the first four slide-forward operations, and the first four slide-backward operations.

Δt	-4	-3	-2	-1	0	1	2	3	4
c_i	0.862	0.021	-0.547	-0.423	0.000	0.867	0.127	-0.466	-0.393

The results indicate that sliding the $\mathbf{b}$ signal forward by one time increment results in the largest normalized cross-correlation with time shift correlation coefficient ($c_1 = 0.867$ at $\Delta t = 1$). This correlation can be seen through visual inspection of the two curves at zero time shift, and then shifting signal $\mathbf{b}$ forward by one step.

...

1.7.7 Continuous Case - Disregard this Section for Now

Finally, let the cross-correlation function $y(t)$ be the sliding inner product of the signal $x(t)$ with the weighting function $w(t)$ such that

$$y(t) = \int_0^{t_f} x(t) w(t - \tau) d\tau, \quad [\text{or?}] \quad y(t) = \int_0^{t_f} x(\tau) w(t - \tau) d\tau, \quad (1.100)$$

which is the filter **slide-forward option**, and

$$y(t) = \int_{-t_f}^0 x(t) w(t - \tau) d\tau \quad [\text{or?}] \quad y(t) = \int_{-t_f}^0 x(\tau) w(t - \tau) d\tau \quad (1.101)$$

⁶See <https://anomaly.io/understand-auto-cross-correlation-normalized-shift/>

is the **slide-backward operation**. The two forgoing operations can be combined to generally be the **sliding dot product operation** as

$$y(t) = \int_{-t_f}^{t_f} x(t) w(t - \tau) d\tau. \quad (1.102)$$

1.7.8 From Deterministic to Stochastic

In this section, we apply some of the notions of expectation, variance, and covariance that we noted in the first sections with the notions of cross-correlation. Up until this point, the signals used in the cross-correlation discussion were tactically normalized to the origin at $\mathbf{0}$. So, expectations and deviations were tactically taken with respect to zero for all times in signals a_i and b_i .

Now, we note the form for correlation applied to stochastic processes (the former will more or less be called a deterministic process, by contrast). In the stochastic process, we will normalize data with respect to an expected value (e.g., mean, in the case of equal likelihood probabilities) as an inhomogeneous datum (different from zero). The mean *could* be zero as a trivial special case, but in general the mean is non-zero.

Stochastic Vector Dot Product

Generally, we substitute $(a_i - \bar{a})$ for (a_i) in the previous definition of vector dot product, appearing in Section 1.7.1. The value $\bar{a}$ is the expected value of $\mathbf{a}$, and in the special case of equal likelihood, $\bar{a}$ is the average value μ_a .

Then, the stochastic vector dot product of three-dimensional signals $\mathbf{a}$ and $\mathbf{b}$ is defined as,

$$\mathbf{a} \cdot \mathbf{b} = (a_i - \bar{a}) (b_i - \bar{b}) \quad (1.103)$$

$$= (a_1 - \bar{a}) (b_1 - \bar{b}) + (a_2 - \bar{a}) (b_2 - \bar{b}) + (a_3 - \bar{a}) (b_3 - \bar{b}) \quad (1.104)$$

$$\text{or generally} = \sum_{i=1}^n (a_i - \bar{a}) (b_i - \bar{b}). \quad (1.105)$$

Stochastic Series Dot Product

Extending this to n -dimensional signals $\mathbf{a}(t)$ and $\mathbf{b}(t)$, the stochastic series dot product becomes,

$$\mathbf{a} \cdot \mathbf{b} = (a_i - \bar{a}) (b_i - \bar{b}) \quad (1.106)$$

$$= (a_0 - \bar{a}) (b_0 - \bar{b}) + (a_1 - \bar{a}) (b_1 - \bar{b}) + (a_2 - \bar{a}) (b_2 - \bar{b}) + \dots + (a_{n-1} - \bar{a}) (b_{n-1} - \bar{b}) \quad (1.107)$$

$$\text{or generally} = \sum_{i=0}^{n-1} (a_i - \bar{a}) (b_i - \bar{b}). \quad (1.108)$$

Stochastic Series Normalized Dot Product

The normalization of the stochastic series dot product arises by comparison of the expected value to its variance. Noting, with the special case of equal likelihood,

$$\sigma^2 = \frac{1}{n} \sum_{i=0}^{n-1} (a_i - \bar{a})^2, \quad (1.109)$$

then,

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=0}^{n-1} (a_i - \bar{a})^2}. \quad (1.110)$$

This allows the stochastic series normalized dot product to be written as

$$\frac{\mathbf{a}}{\|\mathbf{a}\|_2} \cdot \frac{\mathbf{b}}{\|\mathbf{b}\|_2} \triangleq \frac{\frac{1}{n} \sum_{i=0}^{n-1} (a_i - \bar{a}) (b_i - \bar{b})}{\left(\sqrt{\sum_{i=0}^{n-1} (a_i - \bar{a})^2} \right) \left(\sqrt{\sum_{i=0}^{n-1} (b_i - \bar{b})^2} \right)}. \quad (1.111)$$

Some statistical references call this the **zero-normalized cross-correlation**.

Remark 1.7.13. The factor of $\frac{1}{n}$ in (1.111), which arises from the product of two square root variance terms in the denominator, indicates the error *rate*. The error rate, defined in Section 1.8.3, reports error on a per-sample basis.

1.8 Error and Error Rate

1.8.1 Error

The **error** $e(t)$ between two signals $\mathbf{a}(t)$ and $\mathbf{b}(t)$ is defined as the difference between the two signals,

$$\mathbf{e}(t) \triangleq |\mathbf{a}(t) - \mathbf{b}(t)|. \quad (1.112)$$

Note that $\mathbf{e} \geq \mathbf{0} \ \forall \ t$. If $\mathbf{b}$ exactly follows $\mathbf{a}$ at every time step, then the error is zero, as expected.

Example 7.

Let vectors $\mathbf{a}$ and $\mathbf{b}$ be defined as follows,

$$\mathbf{a} = [20, 20, 20, 20] \quad (1.113)$$

$$\mathbf{b} = [18, 18, 18, 18] \quad (1.114)$$

Then the error $\mathbf{e}$ is found to be,

$$\mathbf{e} = [2, 2, 2, 2] \quad (1.115)$$

In this example, assuming $\mathbf{a}$ is the ground truth, $\mathbf{b}$ underestimates $\mathbf{a}$ by 2 at every sample. This is an example of **systemic error**, where there is a consistent offset between the two signals, and an offset adjustment (the offset value of 2 in this case) would eliminate all error between the two signals.

...

1.8.2 Accumulated Error

Next, we would like to have a scalar value to describe the error over the entire sample, which is measured at each sample point and has dimension n .

We define the **L1-norm accumulated error**, $\|\mathbf{e}\|_1$, as the sum of the errors of each of the measurement,

$$\|\mathbf{e}\|_1 \triangleq \sum_{i=0}^{n-1} |e_i| = \sum_{i=0}^{n-1} |a_i - b_i|. \quad (1.116)$$

This is the L1-norm, applied to each error measurement, $e_i = a_i - b_i$. A serious problem with accumulated error is that the error increases the more times the signals are measured.

Example 8.

- In the preceding example, if the signals were measured at only one sample, the accumulated error would be 2.
- Similarly, if two samples were taken, the accumulated error would be 4.
- For three samples, the accumulated error would be 6.
- For the example as stated above, with four samples, the accumulated error is 8.

This pattern shows the limitation of accumulated error: that it is dependent on sample size.

...

We define the **L2-norm accumulated error**, $\| \mathbf{e} \|_2$, as the sum of the errors of

each of the measurement,

$$\| \mathbf{e} \|_2 \triangleq \left(\sum_{i=0}^{n-1} |e_i|^2 \right)^{\frac{1}{2}} = \left(\sum_{i=0}^{n-1} |a_i - b_i|^2 \right)^{\frac{1}{2}}. \quad (1.117)$$

This is the L2-norm, applied to the error measurement, $e_i = a_i - b_i$. The systemic error seen above for the L1-norm, that the error increases with increasing number of measurements, also occurs for the L2-norm. To remedy this effect, normalization by the sample count, called **error rate**, will be used.

1.8.3 Error Rate

The concept of **error rate** is to normalize the accumulated error for any Lp -norm type by the dimension $n = \text{len}(\mathbf{a}) = \text{len}(\mathbf{b})$ of the signals. The resulting metric then

reports error on a *per-measurement basis*, which eliminates the accumulation of error. Specifically, the **L1-norm error rate** is defined as

$$\| \bar{e} \|_1 \triangleq \frac{1}{n} \sum_{i=0}^{n-1} |e_i| = \frac{1}{n} \sum_{i=0}^{n-1} |a_i - b_i|, \quad (1.118)$$

and the **L2-norm error rate** is defined as

$$\| \bar{e} \|_2 \triangleq \frac{1}{n} \left(\sum_{i=0}^{n-1} |e_i|^2 \right)^{\frac{1}{2}} = \frac{1}{n} \left(\sum_{i=0}^{n-1} |a_i - b_i|^2 \right)^{\frac{1}{2}}. \quad (1.119)$$

Notice that the error rate units are [amplitude/sample], hence the rate of the errors in the amplitudes.

Example 9.

Revising the example with vectors **a** and **b**,

$$\mathbf{a} = [20, 20, 20, 20] \quad (1.120)$$

$$\mathbf{b} = [18, 18, 18, 18] \quad (1.121)$$

The **L1-norm error rate** $\| \bar{e} \|_1$ is found to be,

$$\| \bar{e} \|_1 = \frac{1}{4} \cdot (2 + 2 + 2 + 2) = 2. \quad (1.122)$$

The **L2-norm error rate** $\| \bar{e} \|_2$ is found to be,

$$\| \bar{e} \|_2 = \frac{1}{4} \cdot \sqrt{(4 + 4 + 4 + 4)} = 1. \quad (1.123)$$

...

Chapter 2

Dependent and Independent Variables

2.1 Introduction

To come.

2.2 Card Deck Example

Let D denote a regular (no jokers) deck of 52 cards, composed of four suits: hearts, spades, diamonds, and clubs. Each suit is composed of 13 cards: 2, 3, ... 9, 10, J , Q , K , A . We write the probability P of selecting the ace of hearts, $A_{\heartsuit}$, as

$$P(A_{\heartsuit}) = 1/52 = 1.92\%, \quad (2.1)$$

where it is implicit that we are drawing from the full deck D . To be *explicit*, we could equivalently write

$$P(A_{\heartsuit}|D) = 1/52 = 1.92\%, \quad (2.2)$$

which, in words, is “The probability of drawing the ace of hearts **given** that we are drawing from the regular deck of cards is $1/52$.”

We could explore possibilities other than selecting the ace of hearts. Let’s look at the probability of simply selecting an ace, $P(A)$, hearts or otherwise. Since there are four aces in the deck, we write

$$P(A) = P(A|D) = 4/52 = 7.69\%. \quad (2.3)$$

This makes sense. If our chance of drawing an ace of hearts is about 2%, then our chance of just drawing an ace, in general, is about 4 times that, or about 8%.

Next, consider the probability of selecting a heart. We write

$$P(\heartsuit) = P(\heartsuit|D) = 13/52 = 1/4 = 25\%, \quad (2.4)$$

since one-quarter of the cards in the deck are hearts.

Finally, consider the probability of drawing a red card from the deck. We write

$$P(R) = 1/2 = 50\%, \quad (2.5)$$

since half of the cards in the deck are red, and the other half of the cards in the deck are black.

2.2.1 Combining Probabilities

So, does it makes sense to write the probability of drawing an ace of hearts as the *independent* probability of drawing a red card **and** (denoted $\cap$) drawing an ace **and** drawing a heart? Does

$$P(A\heartsuit|D) \stackrel{(?)}{=} P(R) \cap P(A) \cap P(\heartsuit) = \frac{1}{2} \cdot \frac{4}{52} \cdot \frac{1}{4} = \frac{1}{104} \quad (2.6)$$

This answer seems to contradict the $1/52$ result we obtained in Eq. (2.1). What is the explanation?

Before we answer that question, let's look at one other example. Let the probability of drawing a black card, $P(B)$, be denoted as the probability of drawing a “not-red” card, $P(\overline{R})$, where the overbar, $\overline{}$, denotes negation. This makes sense, because the colors, red and black, are binary. If we draw a card that is not red, it must be black. The reason we say “not red” instead of “black” is because we want to keep our focus on the presence or absence of a single variable; in this case, the red variable.

Let's write the probability of drawing a black ace of hearts. Now, already, you should be thinking, “There is no black ace of hearts!” Your intuition tells you that the probability should be zero. But let's see. Let's write the probability of a black ace of hearts, $P(A\spadesuit|D)$, as the independent probability of drawing a black card **and** drawing an ace **and** drawing a heart:

$$P(A\spadesuit|D) \stackrel{(?)}{=} P(\overline{R}) \cap P(A) \cap P(\heartsuit) = \frac{1}{2} \cdot \frac{4}{52} \cdot \frac{1}{4} = \frac{1}{104}, \quad (2.7)$$

which is not zero, the expected result. Again, you may ask, “What is going on here?” The answer is that the three attributes—color, rank, and suit—are being treated as *independent* variables. But, we know that two of those attributes, color and suit, are *not* independent; they are dependent. If we speak of cards with the red color attribute, then we are necessarily talking about hearts and diamonds, and necessarily not talking about spade or clubs. Similarly, if we are speaking of cards with the hearts attribute, we are necessarily talking about the red suit, and necessarily not talking about the black suit.

So, what we have are attributes that are **dependent** on one another. Specifically, color and suit are not independent variables, they are dependent variables. This gives rise to the notion of *conditional variability*.

We note that rank *is* an independent variable. If we speak of rank, we are not necessarily talking about any particular color: red or black. Similarly, if we speak of rank, we are not necessarily speaking of any particular suit: hearts, spade, diamonds, or clubs.

2.2.2 Probability Tree

Going forward, let us generalize variables without hinting at the possibility of conditional variability. To that end, we elect not to color-code hearts and diamonds as red, and we do not color-code the red variable.

- Let the color attribute be described as two variables, red R and black B . Let these two colors be placed into the probability tree (Table 2.1) as red R and not red $\bar{R}$.
- Let the rank attribute be described as 13 variables: 2, 3, ... K , A . Let the binary description of these variables be placed into the probability tree as ace A and not ace $\bar{A}$.
- Let the suit variable be described as 4 variables: hearts H , spades, diamonds, and clubs. Let the binary description of these variables be placed into the probability tree as hearts H and not hearts $\bar{H}$.

The resulting probabilities all make sense with our intuition and correctly calculate the probabilities with the existence of the conditional relationship between the color and suit variables.

Table 2.1: Probability tree to understand card draw probabilities. Note, we keep keep all denominators as specific to the population and avoid reducing the fractions (e.g., $\frac{26}{52}$ is favored to $\frac{1}{2}$) because it is instructive.

variable			probability	interpretation
color	rank	suit		
$P(R) = \frac{1}{2}$	$P(A R) = \frac{2}{26}$	$P(H A R) = \frac{1}{2}$	$P(R \cap A \cap H) = \frac{1}{2} \cdot \frac{2}{26} \cdot \frac{1}{2} = \frac{1}{52} \checkmark$	A of $\heartsuit$
		$P(\overline{H} A R) = \frac{1}{2}$	$P(R \cap A \cap \overline{H}) = \frac{1}{2} \cdot \frac{2}{26} \cdot \frac{1}{2} = \frac{1}{52} \checkmark$	A of $\diamondsuit$
	$P(\overline{A} R) = \frac{24}{26}$	$P(H \overline{A} R) = \frac{1}{2}$	$P(R \cap \overline{A} \cap H) = \frac{1}{2} \cdot \frac{24}{26} \cdot \frac{1}{2} = \frac{12}{52} \checkmark$	$\heartsuit : [2 \dots K]$
		$P(\overline{H} \overline{A} R) = \frac{1}{2}$	$P(R \cap \overline{A} \cap \overline{H}) = \frac{1}{2} \cdot \frac{24}{26} \cdot \frac{1}{2} = \frac{12}{52} \checkmark$	$\diamondsuit : [2 \dots K]$
$P(\overline{R}) = \frac{1}{2}$	$P(A \overline{R}) = \frac{2}{26}$	$P(H A \overline{R}) = \frac{0}{1}$	$P(\overline{R} \cap A \cap H) = \frac{1}{2} \cdot \frac{2}{26} \cdot \frac{0}{1} = 0 \checkmark$	empty set
		$P(\overline{H} A \overline{R}) = \frac{1}{1}$	$P(\overline{R} \cap A \cap \overline{H}) = \frac{1}{2} \cdot \frac{2}{26} \cdot \frac{1}{1} = \frac{2}{52} \checkmark$	A of $\spadesuit$, A of $\clubsuit$
	$P(\overline{A} \overline{R}) = \frac{24}{26}$	$P(H \overline{A} \overline{R}) = \frac{0}{1}$	$P(\overline{R} \cap \overline{A} \cap H) = \frac{1}{2} \cdot \frac{24}{26} \cdot \frac{0}{1} = 0 \checkmark$	empty set
		$P(\overline{H} \overline{A} \overline{R}) = \frac{1}{1}$	$P(\overline{R} \cap \overline{A} \cap \overline{H}) = \frac{1}{2} \cdot \frac{24}{26} \cdot \frac{1}{1} = \frac{24}{52} \checkmark$	$\spadesuit : [2 \dots K]$ $\clubsuit : [2 \dots K]$

- Notice the effect of suit, rows 6 and 8, identifies (makes identical) the probabilities because selecting not hearts $\overline{H}$ given not red $\overline{R}$ gives no new information, thus the 1/1 factor.
- Notice also the effect of suit, rows 4 and 7, nullifies the probabilities because there are zero hearts H that are not red $\overline{R}$, thus the 0/1 factor.

2.3 Coin Toss Example

Consider a fair coin and a fair coin toss (no trick coins, no trick tosses). The probability of a head, $P(H)$ in a toss is 50%. The probability of not head, $P(\overline{H})$, and thus tail, is also 50%.

2.3.1 Combining Probabilities

Consider the results obtain in Eq. (2.6) and Eq. (2.7). Does the probability of getting heads given three coin tosses equal the *independent* probability of tossing a head *and* tossing a second head *and* tossing a third head? Let's see:

$$P(H | \text{three tosses}) \stackrel{?}{=} P(H) \cap P(H) \cap P(H) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}. \quad (2.8)$$

Unlike our suspicions of the results in Eq. (2.6) and Eq. (2.7), our feeling for the results in Eq. (2.8) seem correct. Immediately this result should also bring to mind some sense as to the dependence or independence of each coin toss. Certainly, we know that the results of the second coin toss do not depend on the results of the first coin toss. Similarly, for the third toss relative to the second toss. To formalize these ideas, we create a probability tree (Table 2.2).

2.3.2 Probability Tree

Table 2.2: Probability tree to understand three coin tosses.

variable			probability	interpretation
toss 1	toss 2	toss 3		
$P(H) = \frac{1}{2}$	$P(H H) = \frac{1}{2}$	$P(H H H) = \frac{1}{2}$	$P(H \cap H \cap H) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \checkmark$	3 heads
		$P(\bar{H} H H) = \frac{1}{2}$	$P(H \cap H \cap \bar{H}) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \checkmark$	2 heads, 1 tail
	$P(\bar{H} H) = \frac{1}{2}$	$P(H \bar{H} H) = \frac{1}{2}$	$P(H \cap \bar{H} \cap H) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \checkmark$	2 heads, 1 tail
		$P(\bar{H} \bar{H} H) = \frac{1}{2}$	$P(H \cap \bar{H} \cap \bar{H}) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \checkmark$	1 head, 2 tails
$P(\bar{H}) = \frac{1}{2}$	$P(H \bar{H}) = \frac{1}{2}$	$P(H \bar{H} H) = \frac{1}{2}$	$P(\bar{H} \cap H \cap H) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \checkmark$	2 heads, 1 tail
		$P(\bar{H} \bar{H} H) = \frac{1}{2}$	$P(\bar{H} \cap H \cap \bar{H}) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \checkmark$	1 head, 2 tails
	$P(\bar{H} \bar{H}) = \frac{1}{2}$	$P(H \bar{H} \bar{H}) = \frac{1}{2}$	$P(\bar{H} \cap \bar{H} \cap H) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \checkmark$	1 head, 2 tails
		$P(\bar{H} \bar{H} \bar{H}) = \frac{1}{2}$	$P(\bar{H} \cap \bar{H} \cap \bar{H}) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \checkmark$	3 tails

Unlike the probability results in Table 2.1, the probability results in Table 2.2 are all equal, and the value equals the result found in Eq. (2.8). This indicates the

three variables, toss 1, toss 2, and toss 3, are independent variables, which matches our instinct that the result of any fair coin toss is independent of the result which may have preceded it.

Chapter 3

Binary Classification

3.1 Confusion Matrix

A confusion matrix classifies a population of size n into two columns, with size P and N , described below. The classification of the total population is binary, either as P or N , thus $P + N = n$. Table 3.1 shows the 2-by-2 confusion matrix with the P and N columns.

- The first column is the subset of the population n that is **known positive**, with size P for a condition C .
- The second column is the remaining subset of the population that is **known negative**, with size N for a condition C . Negative status for a condition is denoted with the over set line, $\overline{C}$, to indicate negation of condition C .
- Note $P \leq n$ and $N \leq n$. Typically $P < n$ and $N < n$. Cases of $P = n$ or $N = n$ are degenerate special cases not considered here.

Thus $P = \text{size}(C)$, $N = \text{size}(\overline{C})$, $n = \text{size}(C \cup \overline{C})$, and $0 = \text{size}(C \cap \overline{C})$ since $\emptyset = C \cap \overline{C}$, the binary status of condition C .

- Given a known positive P for a condition C , the P group is divided into two subgroups of true positives TP and false negatives FN , depending on whether or not the test outcome T correctly predicted the truth for the positives in the population. $P = TP + FN$.
- Given a known negative N for condition C , thus $\overline{C}$, the N group is divided into two subgroups of false positives FP and true negatives TN , depending on

whether or not the test outcome T correctly predicted the truth for the negatives in the population. $N = FP + TN$.

- A positive test outcome is denoted T . A negative test outcome is denoted $\bar{T}$.

Table 3.1: The 2-by-2 confusion matrix for the condition C and the result of the test to identify C . Note: the numbers in the table are for the size of each population subset for the worked numerical example.

Total Population $n = 10,000$	Known Condition C $P = TP + FN$	Known Non-Condition $\bar{C}$ $N = FP + TN$
Test Positive T	True Positive TP = 80 “correct affirmation”	False Positive FP = 495 “false alarm” (Type I error)
Test Negative $\bar{T}$	False Negative FN = 20 “miss” (Type II error)	True Negative TN = 9,405 “correct rejection”

3.1.1 Probabilities for the Confusion Matrix

- $P(C)$ is the pre-test probability of having condition C , also known as the **prevalence**.
- $P(\bar{C})$ is then the pre-test probability of not having condition C , thus $\bar{C}$. $P(\bar{C}) = 1 - P(C)$.

When the pre-test probability is combined with a test, the following four outcomes are possible:

- $P(T|C)$ is the probability of a positive test given a known condition. This is a **True Positive** color coded in green.
- $P(T|\bar{C})$ is the probability of a positive test given a known non-condition. This is a **False Positive** color coded in yellow as a warning that perhaps another test should be conducted that might be able to rule out the false positive result.
- $P(\bar{T}|C)$ is the probability of a negative test given a known condition. This is a **False Negative** color coded in orange.

- $P(\bar{T}|\bar{C})$ is the probability of a negative test given a known non-condition. This is a **True Negative** color coded in pink.

Example 10.

Suppose a 44-year-old female has a mammogram test. The radiologist interprets the test result as “suspicious for malignancy.” Assume that the overall probability for any 44-year-old female, regardless of a mammogram test result, to have breast cancer is 1-in-100, or 1% (breast cancer prevalence). Assume the mammogram test is 80% sensitive and 95% specific, from data collected from a population of 10,000 females. Table 3.1 shows the 2-by-2 contingency table for these data.

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It is often useful to show each of the four above-described categories a square, where the area is equal to the population size, as shown in Fig. 3.1.

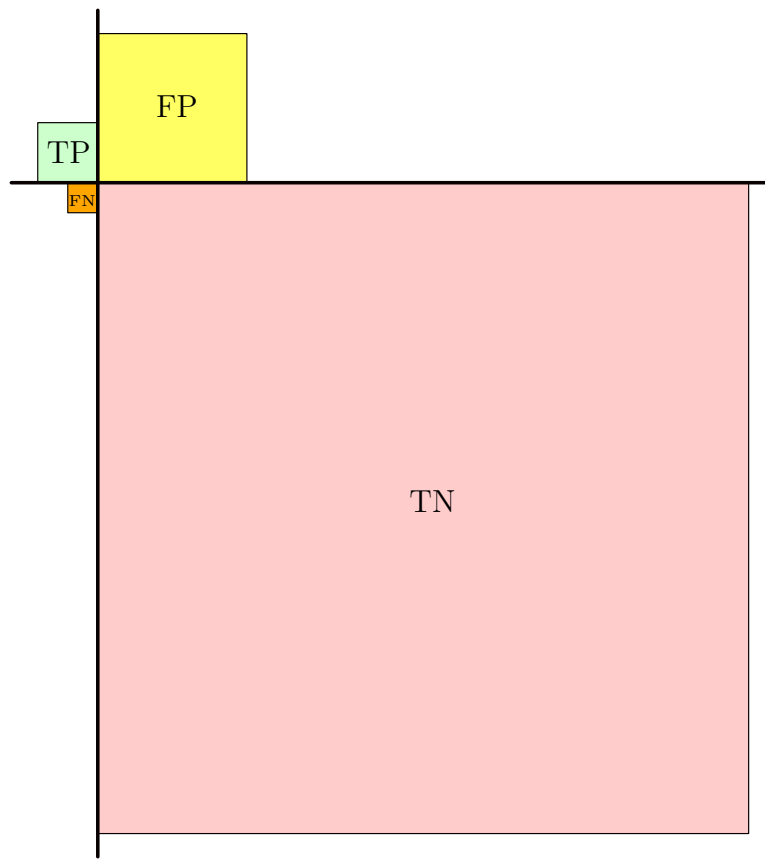
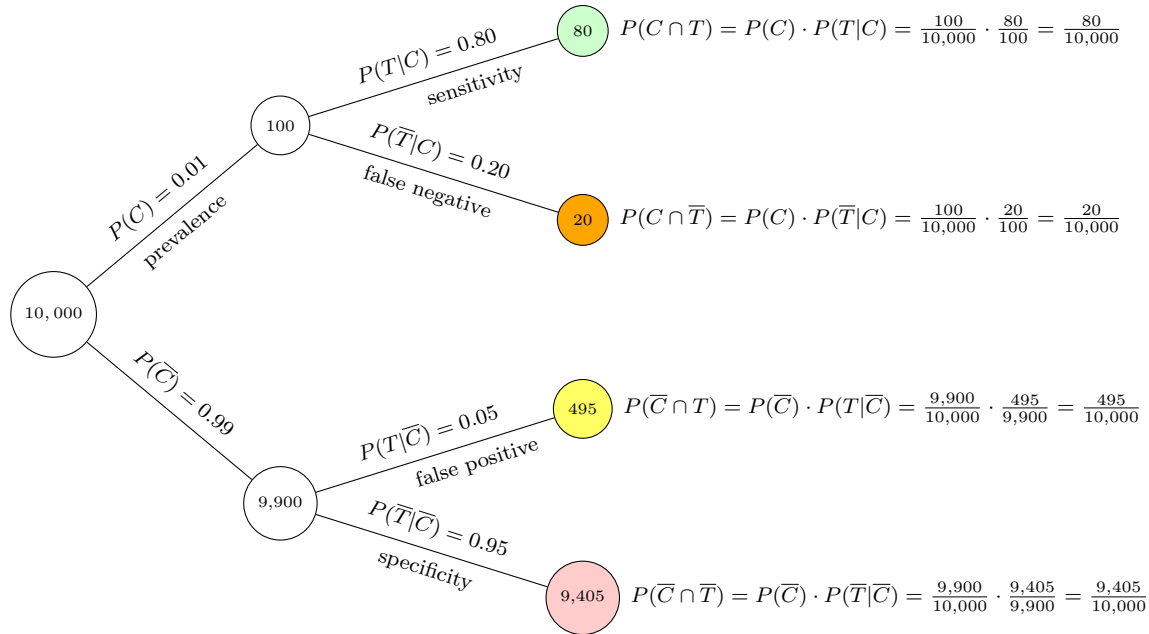


Figure 3.1: Population of true positives ($TP=80$), false positives ($FP=495$), false negatives ($FN=20$), and true negatives ($TN=9,405$) represented by the area of each square.

Figure 3.2 shows the population sorted into a binary tree.

Figure 3.2: Decision tree for mammogram as a test T for breast cancer C .

Now, what is often of interest is the **post-test probability**. For a given person receiving the test, T , what is the probability of having the condition, $P(C|T)$? In fact, all variations can be calculated:

- $P(C|T)$ is the probability of a positive condition given a positive test.
- $P(\bar{C}|T)$ is the probability of a negative condition given a positive test.
- $P(C|\bar{T})$ is the probability of a positive condition given a negative test.
- $P(\bar{C}|\bar{T})$ is the probability of a negative condition given a negative test.

Now, the probability of being positive after testing positive is equal to the probability of being positive and testing positive divided by the probability of testing positive (regardless of being positive or negative for the condition). The probability of testing positive equals the probability of being positive and testing positive *plus* the probability of being negative and testing positive. This is written as

$$P(C|T) = \frac{P(C \cap T)}{P(C \cap T) + P(\bar{C} \cap T)} = \frac{\frac{80}{10,000}}{\frac{80}{10,000} + \frac{495}{10,000}} = 0.139, \quad (3.1)$$

which seems low! A patient, upon hearing this result, that there is a 13.9% probability that she has breast cancer given that she tested positive, does not reassure. The test says she is positive, but the probability that she actually *is* positive given the positive test seems low. How can this be? The large number of false positives, $P(\bar{C} \cap T)$, in the denominator reduces the result for $P(C|T)$. The number of true positives, $P(C \cap T)$, is small compared to the number of false positives. The result in Eq. (3.1) is also referred to as the test's **Positive Predictive Value (PPV)** or **Precision**, described in Section xxx (to come).

Next, consider the remaining combinations. The probability of being negative after testing positive, by reasoning similar to that given above, is written as

$$P(\bar{C}|T) = \frac{P(\bar{C} \cap T)}{P(\bar{C} \cap T) + P(C \cap T)} = \frac{\frac{495}{10,000}}{\frac{495}{10,000} + \frac{80}{10,000}} = 0.861 = 1 - P(C|T) \checkmark. \quad (3.2)$$

Again, we see the effect of false positives, $P(\bar{C} \cap T)$. The probability of being negative even though the test is positive is 86.1%, giving little comfort to the patient who now thinks, because of the test results, that she is positive even though she is negative. The result in Eq. (3.2) is also referred to as the test's **False Discovery Rate (FDR)**.

The probability of being positive after testing negative is

$$P(C|\bar{T}) = \frac{P(C \cap \bar{T})}{P(C \cap \bar{T}) + P(\bar{C} \cap \bar{T})} = \frac{\frac{20}{10,000}}{\frac{20}{10,000} + \frac{9,405}{10,000}} = 0.002 \quad (3.3)$$

This result is reassuring. A patient with a negative test has a 0.2% chance of actually being positive. This occurs because the number of true negatives, $P(\bar{C} \cap \bar{T})$, in the denominator, saturates any false negatives, $P(C \cap \bar{T})$, and drives the result toward zero. The result in Eq. (3.3) is also referred to as the test's **False Omission Rate (FOR)**.

The probability of being negative after testing negative is easily calculated as 1 minus the above result. For explicitness, however, we show

$$P(\bar{C}|\bar{T}) = \frac{P(\bar{C} \cap \bar{T})}{P(\bar{C} \cap \bar{T}) + P(C \cap \bar{T})} = \frac{\frac{9,405}{10,000}}{\frac{9,405}{10,000} + \frac{20}{10,000}} = 0.998 = 1 - P(C|\bar{T}) \checkmark, \quad (3.4)$$

which is also reassuring. Again, the strength of the true negatives, $P(\bar{C} \cap \bar{T})$, overwhelms the effect of the false negatives, $P(C \cap \bar{T})$, in the result. The result in Eq. (3.4) is also referred to as the test's **Negative Predictive Value (NPV)**.